Drainage Strategy

Application reference: 92.73914.FUL
22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP

Proposed drainage

The surface and foul water strategy serves the proposed dwelling at 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP. Building Regulations Part H3 identifies the following order of preference for surface-water disposal: soakaway or infiltration system; watercourse; public sewer.

Percolation testing by the ground investigation consultant identified clayey ground and low infiltration rates, preventing the use of soakaways. There is no watercourse available for discharge because the site is landlocked. Surface water is therefore proposed to discharge to the public surface-water sewer at 2.0 litres/second. Foul water will connect to the public sewer by gravity.

The drained roof area is 300 m² plus 10% urban creep, giving 330 m² (0.033 ha). The drained permeable-paving area is 170 m². The attenuation tank is designed not to flood in the 1-in-100-year event with a 40% climate-change allowance. The proposed tank measures 6.0 m × 12.0 m × 0.4 m deep, with a plan area of 72 m².

Standards and references

The design references BRE Digest 365, Soakaway Design; BS EN 12056-2; Building Regulations Parts H and A; MicroDrainage software; general structural and foundation notes; the Construction (Design and Management) Regulations; and domestic-client roles and responsibilities.

Design responsibilities

The drainage designer has undertaken the Designer role for surface-water and foul-water drainage. The design assumes that an architect has been appointed as Principal Designer and has advised the client of their responsibilities. If an architect has not been appointed, the roles and responsibilities under the CDM regulations normally pass to the principal contractor.

The designer is responsible for providing adequate information about significant design risks. The proposed construction is conventional. The designer considers that the solution reduces foreseeable risk to an acceptable level, that no reasonably avoidable significant risks remain in construction or use, and that the residual risks are normal for this type of project and would be apparent to a competent contractor.

SuDS hierarchy

Surface water should be managed in a way that replicates natural drainage processes as closely as possible. The receptor hierarchy is: infiltration into the ground; a surface-water body or watercourse; a surface-water drain, highway drain or other drainage system; and a combined sewer.

The SuDS considerations refer to CIRIA C753 Table 7.1. Design considerations include runoff rates and volumes, small and large rainfall events, water quality, amenity and biodiversity. Collection mechanisms are identified as point (P), lateral (L) or surface (S).

SuDS components

Rainwater harvesting collects roof or paved-surface runoff for use (P). Green roofs use planted soil layers to slow and store runoff (S). Infiltration systems collect and store runoff for infiltration into the ground (P). Proprietary treatment systems provide subsurface runoff treatment (P).

Filter strips are grass strips that provide sedimentation and filtration as water is conveyed across the surface (L). Filter drains are shallow stone-filled trenches providing attenuation, conveyance and treatment (L). Swales are vegetated channels that convey and treat runoff (L). Bioretention systems are shallow landscaped depressions

that temporarily pond runoff before filtering it (P).

Trees with soil-filled tree pits, planters or structural soils collect, store and treat runoff (P). Pervious pavements allow runoff to pass through structural paving and be stored in the sub-base (S). Attenuation storage tanks provide large below-ground voids that temporarily store runoff before infiltration or controlled release (P). Detention basins are vegetated depressions that store and treat runoff (P). Ponds and wetlands provide permanent pools for runoff treatment and storage (P).

Surface-water drainage design

Infiltration through soakaways is not possible because of the clayey ground and low infiltration rate. There is no watercourse available for discharge because the site is landlocked. Roof runoff is proposed to discharge to the surface-water sewer at 2.0 litres/second, pending a Section 106 agreement with Yorkshire Water.

The 300 m² roof area, with 10% urban creep, gives 330 m² (0.033 ha). A further 170 m² of permeable paving drains to the attenuation tank. The total drained area is 500 m² (0.050 ha). The tank measures 6.0 m × 12.0 m × 0.4 m deep and is designed not to flood in the 1-in-100-year event plus 40% climate change.

The permeable paving serves its own catchment and drains to the surface-water system through 100 mm diameter perforated pipes. It intercepts the first 5 mm of rain and provides bioremediation. The water stored within the paving has not been included in sizing the attenuation tank, making the tank design more conservative.

Storage is offline so that normal rainfall flows do not enter the attenuation tank. This minimises siltation and reduces maintenance. Restricting the small catchment to 2.0 litres/second/hectare would require a very small flow-control opening that would be prone to blockage. A minimum discharge rate of 2.0 litres/second is therefore used.

Rainwater harvesting is not relied upon in the design because it may be removed or inadequately maintained during the life of the development. Individual property owners may install it, but its benefits are excluded from the calculations.

A MicroDrainage model represents the proposed system and the 6.0 m × 12.0 m × 0.4 m attenuation tank. Additional capacity is available in the pipework. The design predicts no flooding in the 1-in-100-year event plus 40% climate change.

Attenuation tank

Roof area: 300 m² plus 10% urban creep = 330 m² (0.033 ha).

Permeable paving: 170 m² (0.017 ha).

Total drained area: 500 m² (0.050 ha).

Invert level: 19.168 mAOD. Top of crates: 19.563 mAOD.

Crate dimensions: 6.0 m × 12.0 m × 0.4 m deep, with 95% voids.

Soakaway investigation

Percolation tests were undertaken to assess a possible soakaway system for surface-water runoff from the proposed residential development. The land investigated forms part of the private garden at 22 Market Lane, Bessacarr, Doncaster, ZZ29 6EP, and comprises an extensive area of grassed soft landscaping. Four trial pits, TP01 to TP04, were excavated using a 1.5-tonne tracked excavator.

The investigation summary records the following dimensions and purposes:

TP01: length 1.3 m; width 0.45 m; depth 1.4 m; hydrogeological assessment.

TP02: length 1.3 m; width 0.45 m; depth 0.95 m; hydrogeological assessment.

TP03: length 1.3 m; width 0.45 m; depth 1.6 m; soakaway testing.

TP04: length 0.4 m; width 0.4 m; depth 0.65 m; soakaway testing.

TP01 and TP03 encountered made ground comprising slightly clayey, very gravelly sand with a low cobble and boulder content, to depths between 0.3 m and 0.8 m below ground level. The gravel comprised fine-to-coarse brick, tile, quartzite and limestone; cobbles and boulders were brick and concrete.

TP02 and TP04 encountered topsoil comprising dark brown clayey gravelly very silty fine-and-medium sand to depths between 0.5 m and 0.6 m below ground level. The gravel comprised fine-to-coarse quartzite. The topsoil or made ground was underlain by firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty clay to the base of each pit. Groundwater was not encountered during the fieldwork.

Testing method and findings

The soakaway test procedure followed the general requirements of BRE DG365 under the supervision of the ground investigation consultant's geologist. Clean potable water was supplied from off site using a water bowser. The vertical sides and base of the pits were trimmed square before testing.

The TP03 test was abandoned after 270 minutes (4.5 hours) because infiltration was too slow for the water level to reach 25% of effective storage depth during the first test. BRE DG365 calls for the pit to be drained on three occasions. The TP04 test was abandoned after 300 minutes (5 hours) for the same reason. The consultant considers the tests representative of the hydrogeological characteristics of the near-surface soils encountered on site.

A traditional soakaway system is considered unlikely to be effective for disposal of surface-water runoff from the development. Groundwater comments are based on observations during the investigation. Observations are affected by the excavation method and rate, and groundwater levels may vary seasonally or for other reasons.

The findings are based on conditions and field tests at the trial-pit locations. Other ground conditions may exist elsewhere on site that were not revealed by the investigation and are not taken into account in the findings.

TP01 exploratory-hole log

From 0.00 m to 0.80 m: MADE GROUND, dark brown slightly clayey very gravelly fine-to-coarse sand with a low cobble and boulder content. Gravel comprises subangular fine-to-coarse brick, quartzite, limestone and tile. Cobbles and boulders are subangular brick and concrete.

From 0.80 m to 1.30 m: firm becoming stiff, indistinctly fissured dark grey mottled brown silty clay, with randomly orientated fissures.

Layer thicknesses: 0.80 m and 0.50 m. End depth: 1.30 m.

Stable conditions; inclination 90 degrees; length 1.40 m; width 0.45 m; Volvo EC18 excavator.

TP02 exploratory-hole log

From 0.00 m to 0.50 m: TOPSOIL, dark brown clayey gravelly very silty fine-and-medium sand. Gravel comprises subrounded fine-to-coarse quartzite.

From 0.50 m to 0.95 m: firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty clay, with randomly orientated fissures.

Layer thicknesses: 0.50 m and 0.45 m. End depth: 0.95 m.

Stable conditions; inclination 90 degrees; length 1.30 m; width 0.45 m; Volvo EC18 excavator.

TP03 exploratory-hole log

From 0.00 m to 0.30 m: MADE GROUND, dark brown slightly clayey very gravelly fine-to-coarse sand with a low cobble and boulder content. Gravel comprises subangular fine-to-coarse brick, quartzite, limestone and tile. Cobbles and boulders are subangular brick and concrete.

From 0.30 m to 1.20 m: firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty clay, with randomly orientated fissures.

From 1.20 m to 1.60 m: firm becoming stiff, indistinctly fissured dark grey mottled brown slightly gravelly silty clay with a low cobble content. Gravel comprises subrounded-to-rounded fine-to-coarse quartzite. Cobbles are rounded quartzite; fissures randomly orientated.

Layer thicknesses: 0.30 m, 0.90 m and 0.40 m. End depth: 1.60 m.

Stable conditions; inclination 90 degrees; length 1.30 m; width 0.45 m; Volvo EC18 excavator.

TP04 exploratory-hole log

From 0.00 m to 0.60 m: TOPSOIL, dark brown clayey gravelly very silty fine-and-medium sand. Gravel comprises subrounded fine-to-coarse quartzite.

From 0.60 m to 0.65 m: firm becoming stiff, indistinctly fissured yellowish brown mottled light grey silty clay, with randomly orientated fissures.

Layer thicknesses: 0.60 m and 0.05 m. End depth: 0.65 m.

Stable conditions; inclination 90 degrees; length 0.40 m; width 0.40 m; Volvo EC18 excavator.

Field test records

Test No. 1 records are provided for TP No. 3 and TP No. 4.

The time entries for both tests are: 0, 1, 2, 3, 4, 5, 10, 15, 20, 25, 30, 45, 60, 75, 90, 105, 120, 150, 180, 210, 240, 270, 300, 360, 420, 480.

TP No. 3 dimensions: width 0.45 m; length 1.4 m; depth 1.3 m. Standing-water readings, in recorded order (mm below ground level): 527, 527, 529, 532, 534, 535, 535.

TP No. 4 dimensions: width 0.4 m; length 0.4 m; depth 0.65 m. Standing-water readings, in recorded order (mm below ground level): 198, 201, 204, 209, 209, 213, 214, 215, 215, 216, 218.

Network calculation parameters

Drainage design reference: 4555. Model file: 1 IN 100Y+40%.MDX. Software: Innovyze network calculations. Method: Modified Rational Method.

Pipe sizes: STANDARD. Manhole sizes: STANDARD. Rainfall model: FEH. Return period: 2 years. Data type: point. Maximum rainfall: 50 mm/hour. Maximum time of concentration: 30 minutes.

Foul sewage: 0.000 litres/second/hectare. Volumetric runoff coefficient: 1.000. PIMP: 100%. Additional flow/climate change: 0%.

Minimum backdrop height: 0.200 m. Maximum backdrop height: 1.500 m. Minimum design depth for optimisation: 1.200 m. Minimum velocity for auto design only: 1.00 m/second. Minimum optimisation slope: 1:500. Designed with level inverts.

Time-area entries: 0-4 minutes, 0.038 ha; 4-8 minutes, 0.012 ha. Total contributing area: 0.050 ha. Total pipe volume: 0.750 m³.